WHITEPAPER · BRAZIL 2026

Between the Rounds

Estimating how Brazilian voters moved from the first to the second round of the 2026 presidential election.

Rennan Chagas · Working paper v1.0 · October 30, 2026 · Exploratory results · Short version (blog post)

SUMMARY

exploratory

97.3% · 95.1%
of Lula's and Flávio's first-round voters voted for them again

3.81M vs 3.82M
voters of eliminated candidates who went to Lula versus Flávio: an almost even split

+4.8 million
net advantage for Lula among round-1 non-voters and blank/null voters: more than his whole 4.4 million margin

1.3% → 11.5%
typical share of voters misplaced in simulated elections, from the friendliest to the hardest scenario

1. Introduction

What this paper does not do.

2. Data

2.1 Official ballot-box results

boletim de urna

seção

exactly

2.2 Matching ballot boxes across rounds

2.3 Nine kinds of first-round voter, five second-round choices

2.4 Voters who changed ballot box

temporary transfer

2.5 Opinion polls

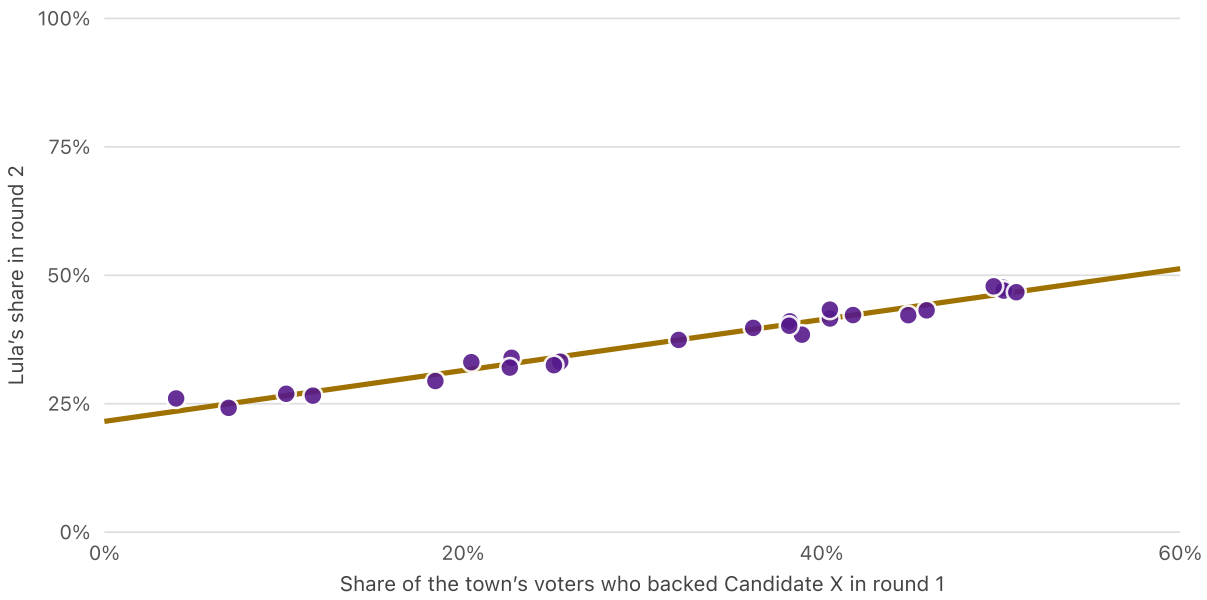
3. Method, in plain language

3.1 The core idea

ecological inference

Figure 1 · The core idea, and where it breaks

Each dot is an imaginary town; the line is the best straight-line summary.



Real share of X-voters who chose Lula	What the line says	Error
70%	**71%**	**1 pp**

Made-up data. In this world, 70% of Candidate X's voters really chose Lula, and 22% of everyone else did.

ecological bias

3.2 The model

- **183 areas**

same chance

-

- *overdispersion*

-

3.3 Making the books balance

iterative proportional fitting

3.4 People who changed ballot box

3.5 Quantifying uncertainty

- **Hard limits.**

- **95% range from resampling.**
  bootstrap

3.6 Blending in polls (sensitivity analysis)

For the curious: the model in a few lines

s x_{si} i y_{sj}
j p_{ij} i j

```
mean: μ
```

p_i

$f \cdot x_s P +$

$e \cdot q$ f e q

→

`docs/population/model_specification.md`

4. Results

Where

Registration-move assumption

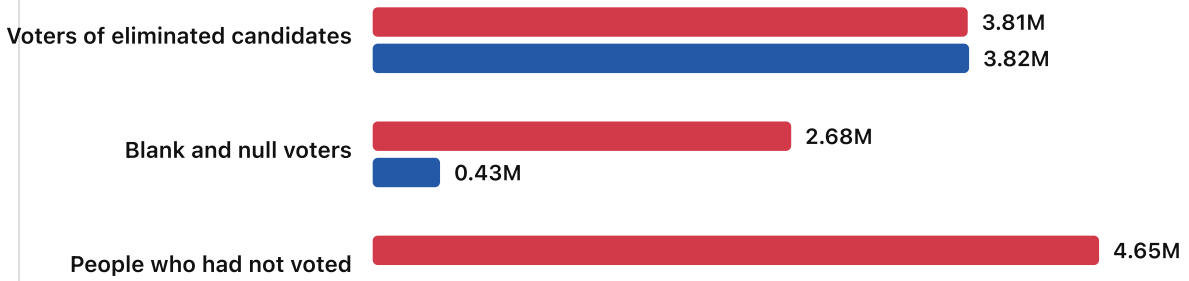
4.1 The big picture

Figure 2 · Where each group of voters ended up

Left: round-1 choice ▒▒ t: round-: ▒hoice ▒▒? Ô▒▒▒ ▒▒▒▒ ▒▒ ✗Æ e▒o▒▒

Figure 3 · Where each finalist's second-round votes came from

By type of first-round voter. Switch to "Net advantage" to see how the final margin is built.



Brazil (whole country): Lula kept 97.3% of his round-1 voters; Flávio kept 95.1% of his.

The four net bars add up to the final margin between the finalists.

- **Eliminated candidates' voters**

- **Blank and null voters**
- **Round-one non-voters**

against

4.3 The full table

Figure 4 · Transition matrix

Each row is a first-round group; each column a second-round choice.

	Lula	Flávio Bolsonaro	Blank	Null	Did not vote	Total
Lula	97.3% 52,417,073	0.15% 82,468	0.011% 5,695	0.12% 64,903	2.4% 1,285,687	53,879,538
Flávio Bolsonaro	1.2% 674,805	95.1% 53,338,561	0.14% 77,424	0.097% 54,533	3.5% 1,935,879	56,104,503
Augusto Cury	51.6% 1,778,107	34.8% 1,200,720	5.0% 170,747	5.8% 199,696	2.8% 97,882	3,448,569
Renan Santos	35.8% 957,595	37.9% 1,012,524	4.5% 119,938	12.0% 320,504	9.9% 263,879	2,675,887
Ronaldo Caiado	35.2% 916,811	50.3% 1,310,017	5.0% 129,627	5.4% 139,449	4.2% 108,138	2,605,148
Other candidates	27.1% 158,787	50.6% 296,729	4.2% 24,873	11.5% 67,227	6.7% 39,149	587,143
Blank	49.5% 1,138,219	7.3% 168,829	36.6% 841,053	2.1% 47,792	4.5% 104,212	2,300,798
Null	42.0% 1,542,399	7.2% 262,642	0.84% 30,683	46.3% 1,702,413	3.7% 135,039	3,674,249
Did not vote	13.9% 4,652,654	6.3% 2,123,311	0.085% 28,526	0.37% 124,046	79.3% 26,528,768	33,469,244
Total	64,256,232	59,826,041	1,430,496	2,723,305	30,509,005	158,745,079

Brazil (whole country): 158,745,079 registered voters; the model was fitted in 183 areas (1 did not fully converge). Rows add up to 100% (ignoring about 0.04% of voters whose registration moved). Hover or tap a cell for the 95% range.

Cell color is proportional to the share of the row. Hover or tap a cell for the 95% range and the hard limits.

4.4 Geography

Figure 5 · Geography of the transitions

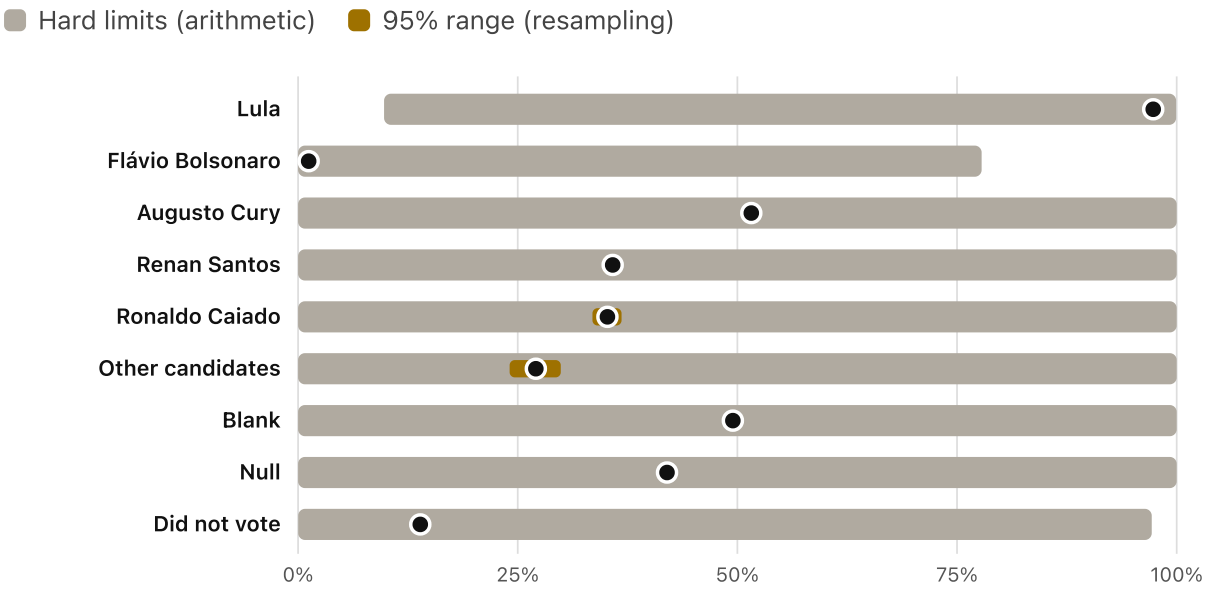
A tile map of the 26 states

5. How reliable are these estimates?

5.1 The ranges

Figure 6 · Estimates, 95% ranges and hard limits

Share of each first-round group that went to the selected destination.



Brazil (whole country): 158,745,079 registered voters; the model was fitted in 183 areas (1 did not fully converge).

Grey: hard limits from the official totals alone. Gold: 95% range from resampling areas (1,000 draws). Dot: best estimate. Ranges exist for the base case only.

5.2 Tests on simulated elections

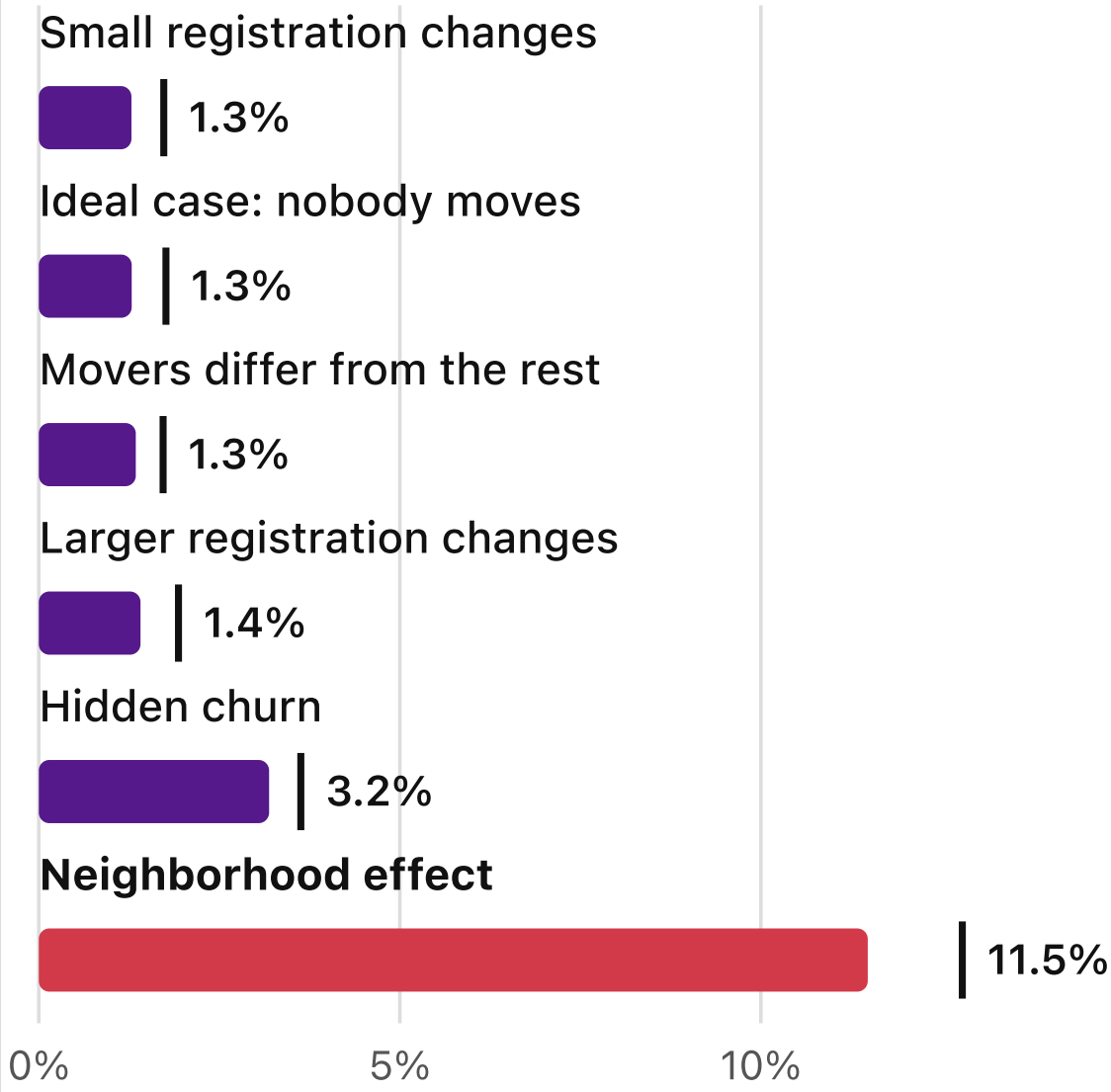
- **Ideal case:**
- **Registration changes (small, larger, selective):**

- **Hidden churn:**
- **Neighborhood effect:**

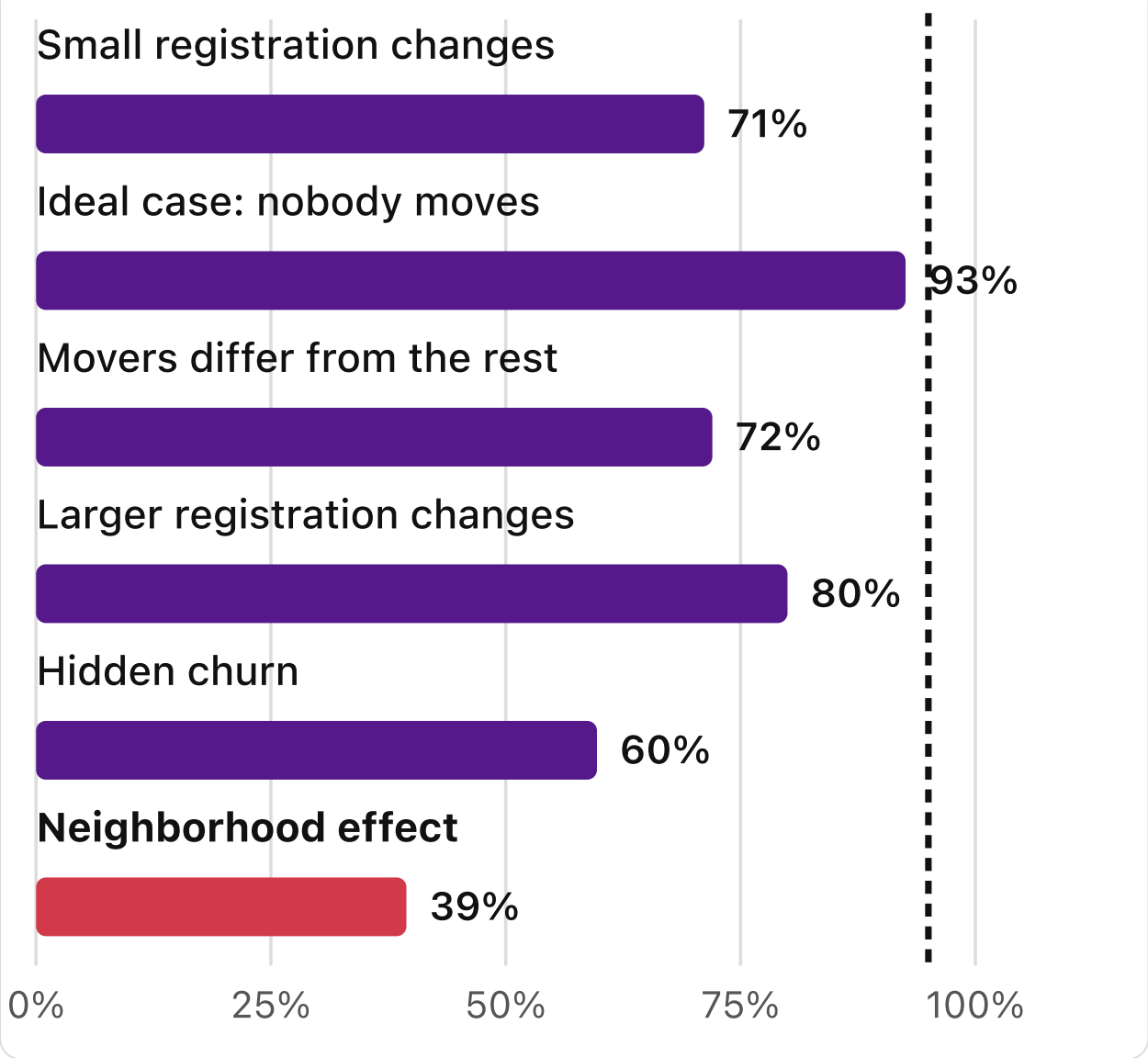
Figure 7 · Accuracy on simulated elections

Mean error and how often the 95% range contained the truth.

Average share of voters in the wrong box



Share of "95% ranges" that contained the truth (target: 95%)



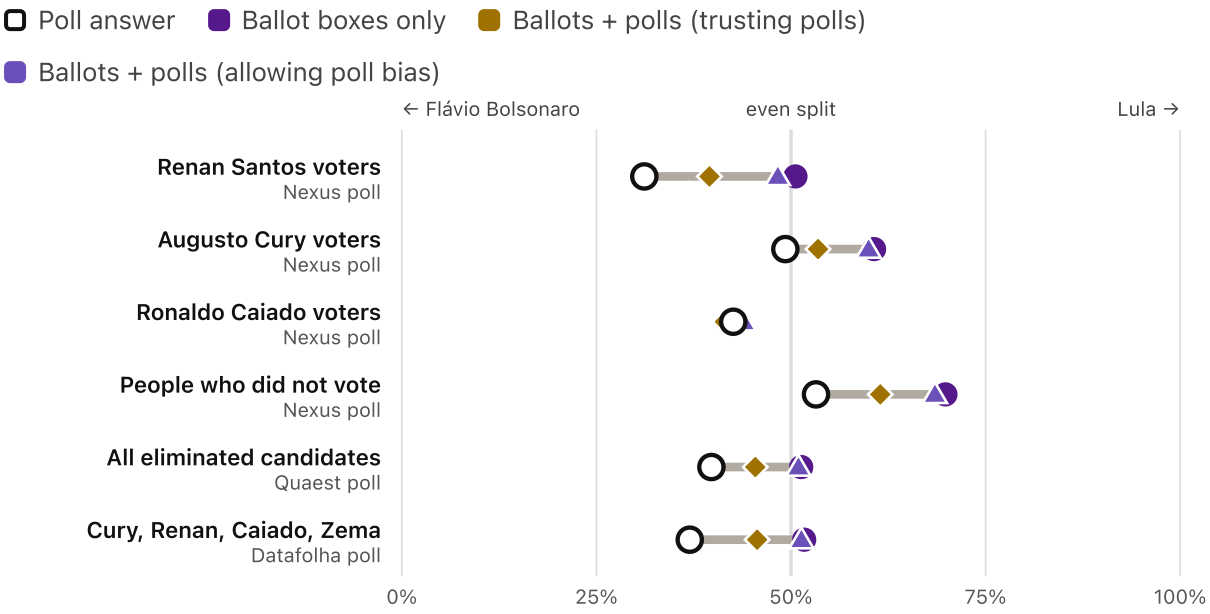
The red bar marks the scenario where voting habits vary with local make-up. Black ticks on the left panel mark the 95th percentile of the error.

Scenario	Typical error	Bad-day error (worst 5%)	Typical error if boxes with changed rosters are dropped	Share of "95% ranges" that held	Fits fully converged
Small registration changes	1.3%	1.7%	1.5%	71%	63%
Ideal case: nobody moves	1.3%	1.8%	1.3%	93%	30%
Movers differ from the rest	1.3%	1.7%	1.6%	72%	63%
Larger registration	1.4%	1.9%	6.7%	80%	95%

Scenario	Typical error	Bad-day error (worst 5%)	Typical error if boxes with changed rosters are dropped	Share of “95% ranges” that held	Fits fully converged
changes					
Hi					

Figure 8 · Ballot boxes versus polls

Among people who chose Lula or Flávio, the share who chose Lula, by first-round group.



Polls measure vote intention shortly before the vote from small samples; the ballot-box method measures the actual result with model assumptions. Neither is ground truth.

5.4 Sensitivity to assumptions

Figure 9 · Do the headline numbers depend on our assumptions?

Brazil-wide estimates under the six alternative assumptions about registration moves and model-area size.

	Base case	Base, larger areas	Fewest moves	Fewest, larger areas	More moves	Many moves
Lula voters who stayed with Lula	97.3%	97.4%	97.3%	97.4%	97.7%	98.0%
Flávio voters who stayed with Flávio	95.1%	95.3%	95.1%	95.3%	95.6%	96.0%
Eliminated candidates' voters who chose Lula*	49.9%	50.4%	49.9%	50.4%	50.5%	51.2%
Blank/null voters who chose Lula*	86.1%	86.1%	86.2%	86.2%	85.8%	85.5%
Non-voters of round 1 who voted in round 2	20.7%	20.6%	20.7%	20.6%	19.9%	19.2%
Returning non-voters who chose Lula*	68.7%	69.2%	68.7%	69.2%	69.2%	69.7%

*Among those who chose Lula or Flávio. The numbers hardly move: these assumptions are not what limits the analysis.

these

5.5 Numerical health

6. Limitations

No individual-level data.

The neighborhood effect.

Ranges are too narrow.

Moves between ballot boxes.

Small places.

Small candidates.

Polls.

Not a replication.

Not peer reviewed.

7. Reproducibility

```
 python3 -m venv .venv && .venv/bin/python -m pip install -r requirements.lock.txt
.venv/bin/python -m src.pipeline all          # baseline analysis
.venv/bin/python -m src.population.pipeline   # extension used in this paper
.venv/bin/python site_src/extract_data.py     # data behind these pages
```

8. Glossary

Ballot box / section (*seção*)

Blank / null vote

Abstention / "did not vote"

Ecological inference

Ecological bias / neighborhood effect

Bootstrap / "95% range from resampling"

Hard limits

Temporary transfer

Iterative proportional fitting

9. References and sources

1. Brown, P. J., & Payne, C. D. (1986). Aggregate data, ecological regression, and voting transitions. *Journal of the American Statistical Association*, 81(394), 452–460.
2. Forcina, A., & Pavía, J. M. (2026). Estimating Vote Transition Counts in Ecological Inference. *Statistics and Public Policy, 13*(1), Article 2654396. https://doi.org/10.1080/2330443X.2026.2654396.
3. Klima, A., Schlesinger, T., Thurner, P. W., & Küchenhoff, H. (2017). Combining aggregate data and exit polls for the estimation of voter transitions. *Sociological Methods & Research*. doi:10.1177/0049124117701477
4. Greiner, D. J., & Quinn, K. M. (2010). Exit polling and racial bloc voting: Combining individual-level and R×C ecological data. *The Annals of Applied Statistics*, 4(4), 1774–1796. doi:10.1214/10-AOAS353
5. Tribunal Superior Eleitoral. *Dados Abertos: Resultados 2026; Eleitorado 2026*. dadosabertos.tse.jus.br

6. Quaest (BR-09158/2026), Nexus/BTG (BR-08234/2026) and Datafolha (BR-09540/2026) presidential polls, October 2026, as registered with the TSE.